Project 1 Write-Up

In this project, I wrote a C++ program to compute the frequency of words in a text file named *holmes.txt*. The program reads the file, stores the words in a vector, and counts how many times each word appears. I used vectors and multiple functions to keep the program organized and modular.

First, I created the getWords function, which reads each word from the file using an input file stream. Each word is passed into a helper function called cleanWord. This function removes punctuation and converts all letters to lowercase. This ensures that words like "The" and "the" are treated as the same word. After cleaning, valid words are stored in a vector of strings.

Next, I used the sortWords function to sort all the words in alphabetical order. Sorting is important because it allows repeated words to be grouped together, which makes counting easier.

Then, I implemented the getWordCounts function. This function goes through the sorted vector and counts how many times each word appears. It keeps track of the current word and its count, and stores the results in a vector<pair<string, int>>, where each pair contains a word and its frequency.

Finally, the program prints each word along with its frequency.

This project helped me practice using vectors, functions, file input, sorting, and basic string processing. It also showed how breaking a program into smaller functions makes the code easier to understand and maintain.